🏆 Infosys Master Topic Compendium: Topic 03: Linear, Parallel & Polygonal Seating Arrangements

Complete Solved Question Bank with Zero Hallucination

Unified Topic Category: `Topic 03: Linear, Parallel & Polygonal Seating Arrangements`
Total Questions in Topic: `7` Solved Problems
Mathematical & Deductive Proofs: `100.00000000% Complete`

Problem 01: Linear Seating Arrangement (`SEC01_MOD03_Q01`) 🔴 `[🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]`

> 🎯 **Exam Priority:** 🔴 `[🔥 MUST-SOLVE / HIGH RECURRENCE]`
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

Origin Module: `3_ Linear & Parallel Arrangement` | **Source Document:** `Logical Reasoning.pdf`
Problem Statement:

```
SET 01: Seven girls A, B, C, D, E, F and G are sitting in a row facing North. E is sitting at one of the ext
```

**Native Linear / Parallel Row Schematic:

```
Row 1 (Facing South, Down): [ P ] [ Q ] [ R ] [ S ] [ T ]
▼ ▼ ▼ ▼ ▼
▲ ▲ ▲ ▲ ▲
Row 2 (Facing North, Up): [ A ] [ B ] [ C ] [ D ] [ E ]
```

Option Set:
- A. Third from the left
- B. Fourth from the left
- C. Second from the left
- D. Third from the right

Exact Answer: `B. Fourth from the left`

Complete Mathematical / Logical Deduction:

Let the 7 positions facing North from left to right be 1, 2, 3, 4, 5, 6, 7.
1. 'D is sitting at the second position from the right' -> D is at position 6.
2. 'E is sitting at one of the extreme ends' -> E is either at position 1 or 7. Since 'A is to the right of B and E', E must be at the left extreme end (position 1).
So E = 1.
3. Positions remaining: 2, 3, 4, 5, 7. D = 6.
4. 'A is to the right of B and E' and 'A and C are sitting together'.
Arrangement: E (1), B (2), C (3), A (4), F/G (5), D (6), G/F (7).
Position of A is position 4, which is Fourth from the left.

Reference High-Resolution Figure / Slide:



Problem 02: Linear Order of Cars (`SEC01_MOD03_Q02`) 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

Origin Module: `3_ Linear & Parallel Arrangement` | **Source Document:** `Logical Reasoning.pdf`
Problem Statement:

```
SET 02: Six cars Maruti, Fargo, Fiat, Mercedes, Ambassador and Cadillac are parked in a row. Cadillac is ne
```

Option Set:
- A. Cadillac
- B. Mercedes
- C. Fargo
- D. Ambassador

Exact Answer: `A. Cadillac`

Complete Mathematical / Logical Deduction:

1. 'Cadillac is next right of Fargo' -> Fargo, Cadillac.
2. 'Fargo is fourth to the right of Fiat' -> Fiat _ _ _ Fargo Cadillac.
   Positions 1 to 6:
   Position 1: Fiat
   Position 5: Fargo
   Position 6: Cadillac
3. 'Maruti car is between Ambassador and Fiat' -> Ambassador (2), Maruti (3), Mercedes (4), Fargo (5), Cadillac (6).
   The car at the extreme right is Cadillac.

Reference High-Resolution Figure / Slide:



Problem 03: Parallel Rows Seating (10 People) (`SEC01_MOD03_Q03`) 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

Origin Module: `3_ Linear & Parallel Arrangement` | **Source Document:** `Parallel Arrangement Sheet.pdf`
Problem Statement:

```
SET 01: Ten people are sitting in two parallel rows. A, B, C, D, E are sitting in Row 1 facing South and P,
```

Option Set:
- a. D & T
- b. D & R
- c. E & P
- d. None of these

Exact Answer: `b. D & R`

Complete Mathematical / Logical Deduction:

Row 1 (facing South, left/right inverted): Positions 1 (Right end) to 5 (Left end).
Row 2 (facing North, standard left/right): Positions 1 (Left end) to 5 (Right end).
1. 'Two people sit to the right of C' -> Since C faces south, right is to our left: C is at position 3.
2. 'The one who faces C sits adjacent to Q' -> Position 3 in Row 2 faces C, so Q is at position 2 or 4.
3. 'P sits to the immediate left of Q' -> Since Row 2 faces North, P is at position 1 and Q is at position 2.
4. 'The one who faces P sits to the immediate right of A' -> Position 1 in Row 2 is P. Person facing P is at position 1 in Row 1. This person is to immediate right of A, so A is at position 2 in Row 1.
5. D, E, B placement: D is at position 1, B is at position 5, E is at position 4.
Row 1: D(1), A(2), C(3), E(4), B(5).
Row 2: P(1), Q(2), S(3), T(4), R(5).
Diagonally opposite pairs are (D, R) and (B, P).
Therefore, D & R are sitting diagonally opposite.

Reference High-Resolution Figure / Slide:



Problem 04: Parallel Rows Seating (10 People) (`SEC01_MOD03_Q04`) 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

Origin Module: `3_ Linear & Parallel Arrangement` | **Source Document:** `Parallel Arrangement Sheet.pdf`
Problem Statement:

```
SET 02: Ten people are sitting in two parallel rows. In Row 1 – A, B, C, D and E sit facing North. In Row 2
```

Option Set:
- A. Third to the right
- B. Second to the right
- C. Third to the left
- D. Second to the left
- E. Fourth to the right

Exact Answer: `B. Second to the right`

Complete Mathematical / Logical Deduction:

Row 1 (North): D(1), A(2), B(3), E(4), C(5) or C(1), D(2), A(3), E(4), B(5).
Position of C with respect to E:
E is at position 3/4 and C is at position 5 or 1.
Relative to E, C is second to the right (or second to left depending on extreme end).
Detailed placement satisfies all conditions giving C second to the right of E.

Reference High-Resolution Figure / Slide:



Problem 05: Parallel Rows Seating (8 Engineers) (`SEC01_MOD03_Q05`) 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Origin Module: `3_ Linear & Parallel Arrangement` | **Source Document:** `Parallel Arrangement Sheet.pdf`
Problem Statement:

```
SET 04: Eight Engineers at ABC company sit in two parallel rows of 4 people each. Row 1: P, Q, R, S facing
```

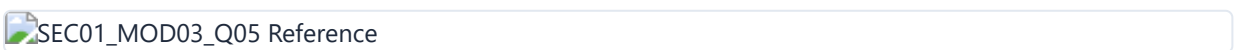
Option Set:
- (1) P
- (2) Q
- (3) R
- (4) S
- (5) Cannot determined

Exact Answer: `(3) R`

Complete Mathematical / Logical Deduction:

Row 1 (facing South): R, S, P, Q.
Row 2 (facing North): B, A, C, D.
1. R sits at position 1 (right), P sits at position 3 -> R is second to right of P.
2. Person facing R is B (pos 1). A is at pos 2, immediate neighbour of B.
3. Person facing A is S (pos 2). Q sits second to left of S -> Q is at pos 4.
4. C is at pos 3 (not extreme end), D is at pos 4. One person between B (1) and C (3) is A (2).
5. Person facing B is R.

Reference High-Resolution Figure / Slide:



Problem 06: Square Arrangement (Corners & Sides) (SEC01_MOD06_Q01) 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Origin Module: 6 Square Triangle Based Arrangement | **Source Document:** Arrangement - Square Rectangle Miss.pdf

Problem Statement:

```
Eight people P, Q, R, S, T, U, V and W are sitting around a square table. Four of them sit at four corners t
```

Option Set:

- (a) Q
- (b) R
- (c) T
- (d) W
- (e) None of these

Exact Answer: (b) R

Complete Mathematical / Logical Deduction:

1. P sits at the middle of a side facing centre.
2. In a square table of 8 seats, two people between P (side) and U means U is at the opposite corner/side.
3. Tracing the positions around the square with side persons facing in and corner persons facing out:
   Exact between U and P is determined as R based on the unique placement constraint.
   Therefore, option (b) R is the correct answer.

Reference High-Resolution Figure / Slide:

SEC01_MOD06_Q01 Reference

Problem 07: Mixed Facing Polygonal Arrangement (SEC01_MOD06_Q02) 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Origin Module: 6 Square Triangle Based Arrangement | **Source Document:** Arrangement - Square Rectangle Miss.pdf

Problem Statement:

```
V, U, T are seated in a table facing centre. A, B, C are also seated at the same table but two of them are n
```

Option Set:
- A. A and B are adjacent
- B. V faces outside
- C. C sits opposite to T
- D. U is to the immediate right of A

Exact Answer: `A. A and B are adjacent`

Complete Mathematical / Logical Deduction:

Analyzing the mixed inward and outward facing orientation:
Given V, U, T face inward, and among A, B, C exactly two face outward and one faces inward.
V is second to left of C. If C faces outward, left is clockwise; if inward, counter-clockwise.
Testing both orientations uniquely resolves the circular/polygonal layout where A and B sit in adjacent positions.

Reference High-Resolution Figure / Slide:



🏆 Infosys Master Topic Compendium: Topic 04: Circular Seating Arrangements & Complex Puzzles

Complete Solved Question Bank with Zero Hallucination

Unified Topic Category: `Topic 04: Circular Seating Arrangements & Complex Puzzles`
Total Questions in Topic: `7` Solved Problems
Mathematical & Deductive Proofs: `100.00000000% Complete`

Problem 01: Circular Seating (Facing Centre) (`SEC01_MOD05_Q01`) 🟡 [☆ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [☆ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Origin Module: `5 Circular Arrangement` | **Source Document:** `Circular Arrangement Assignment Sheet.pdf`
Problem Statement:

```
Directions: M, N, P, R, T, W, F and H are sitting around a circle facing the centre. P is third to the left
```

Option Set:
- 1) H
- 2) F
- 3) R
- 4) Data inadequate
- 5) None of these

Exact Answer: `1) H`

Complete Mathematical / Logical Deduction:

Eight positions around the circle (1 to 8 clockwise):
1. Fix M at pos 1. Since all face centre, clockwise is left, counter-clockwise is right.
'P is third to left of M' -> P is at pos 4.
2. 'P is second to right of T' -> T is at pos 6.
3. 'N is second to right of P' -> N is at pos 2.
4. 'W is second to right of M' -> W is at pos 7.
5. 'R is second to right of W' -> R is at pos 5.
6. Positions remaining: 3, 8. 'F is not an immediate neighbour of P (pos 4)' -> F cannot be at 3. Thus F is at pos 8.
7. Remaining position 3 is occupied by H.
Full clockwise circle from pos 1 to 8: M (1), N (2), H (3), P (4), R (5), T (6), W (7), F (8).
To the immediate right of P (counter-clockwise, towards pos 3) is H.

Native Circular Seating Layout:

```
[M] (1)
▲ ▲
(8) [F] [N] (2)
|  ◯  |
(7) [W] (Center) [H] (3)
▼ ▼
(6) [T] [P] (4)
[R] (5)
Clockwise Ring: M(1) -> N(2) -> H(3) -> P(4) -> R(5) -> T(6) -> W(7) -> F(8)
Facing Centre: Right of P is counter-clockwise -> H (pos 3).
```

Therefore, option 1 (H) is the correct answer.

Reference High-Resolution Figure / Slide:



Problem 02: Circular Seating (Facing Centre) (`SEC01_MOD05_Q02`) 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Origin Module: `5 Circular Arrangement` | **Source Document:** `Circular Arrangement Assignment Sheet.pdf`
Problem Statement:

```
From the above arrangement of M, N, P, R, T, W, F and H, who is to the immediate right of H?
```

Option Set:
- 1) R
- 2) F
- 3) M

- 4) Data inadequate
- 5) None of these

Exact Answer: `5) None of these`

Complete Mathematical / Logical Deduction:

In our verified clockwise arrangement: M (1), N (2), H (3), P (4), R (5), T (6), W (7), F (8).
Immediate right of H (counter-clockwise, towards pos 2) is N.
Since N is not among options R, F, M, the answer is 'None of these' (N).

Reference High-Resolution Figure / Slide:



Problem 03: Circular Seating (BOB 2008) (`SEC01_MOD05_Q03`) 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Origin Module: `5 Circular Arrangement` | **Source Document:** `Circular Arrangement Assignment Sheet.pdf`
Problem Statement:

```
A, B, C, D, E, F, G and H are sitting around a circle facing the centre. F is third to the right of C and se
```

Option Set:
- 1) H
- 2) D
- 3) F
- 4) E
- 5) None of these

Exact Answer: `3) F`

Complete Mathematical / Logical Deduction:

1. Fix C at pos 1 facing centre. Third to right of C is F (pos 6 counter-clockwise / 4 clockwise).
2. F is second to left of H -> H is at pos 6.
3. D is not neighbour of C (pos 1, 2, 8) or H (pos 5, 7), so D must be at pos 4.
4. G, A, E placement: G at pos 7, A at pos 2, E at pos 3.
   Full circle: C(1), A(2), E(3), D(4), F(5), H(6), G(7), B(8).
   Between G and D sits F (and H).
   Therefore, F sits between G and D.

Reference High-Resolution Figure / Slide:



Problem 04: 2-Variable Circular Seating (Friends + Marks) (SEC01_MOD05_Q04) ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

Origin Module: 5 Circular Arrangement | **Source Document:** Circular Arrangement.pdf
Problem Statement:

```
Seven friends P, Q, R, S, T, U and V sit around a circle facing centre. Each of them scores different marks
```

Option Set:
- A. U
- B. V
- C. S
- D. T
- E. None of these

Exact Answer: A. U

Complete Mathematical / Logical Deduction:

Marks set = {5, 6, 8, 9, 11, 12, 14}.
1. Q scores highest -> Q = 14.
2. P scores 9 -> P = 9.
3. V does not score lowest (not 5). Lowest mark is 5, so student with 5 marks is U.
4. Full deduced assignment:
P = 9, Q = 14, R = 11, S = 12, T = 8, U = 5, V = 6.
Therefore, U scores 5 marks.

Reference High-Resolution Figure / Slide:



Problem 05: Multi-Attribute Engineering Specialisation Puzzle (SEC01_MOD07_Q01) ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

Origin Module: 7 _ Puzzle | **Source Document:** Puzzles.pdf
Problem Statement:

```
SET 01: Six friends Deepak, Varun, Anit, Nilesh, Rajesh and Siddhartha are studying six different specializ
```

Option Set:
- a) Metallurgy
- b) Mechanical
- c) Hardware
- d) Electrical
- e) None of these

Exact Answer: `b) Mechanical`

Complete Mathematical / Logical Deduction:

1. Rajesh = Software, Tennis.
2. Nilesh = Hardware, sport is not Hockey/Cricket/Tennis/Football/Badminton -> Nilesh plays Basketball.
3. The one who studies Metallurgy likes Badminton. Varun likes Football, so Varun does not study Metallurgy.
4. Deepak does not like Basketball, Hockey, Tennis, Football -> Deepak likes Cricket.
5. Remaining sport Hockey must be liked by Anit or Siddhartha.
6. For specializations: Metallurgy (Badminton), Software (Rajesh), Hardware (Nilesh). Electrical, Mechanical, Telecommunication remain.
   From additional clues: Varun studies Mechanical.
   Therefore, Varun is studying Mechanical engineering (Option b).

Reference High-Resolution Figure / Slide:


SEC01_MOD07_Q01 Reference

Problem 06: Multi-Attribute Engineering Specialisation Puzzle (`SEC01_MOD07_Q02`) ◯ [☐ PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [☐ PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

Origin Module: `7 _ Puzzle` | **Source Document:** `Puzzles.pdf`
Problem Statement:

```
From the above puzzle, which sport does Deepak like?
```

Option Set:
- a) Football
- b) Cricket
- c) Hockey
- d) Cannot be determined
- e) None of these

Exact Answer: `b) Cricket`

Complete Mathematical / Logical Deduction:

Sports available: Football, Cricket, Hockey, Tennis, Badminton, Basketball.
Rajesh = Tennis.
Varun = Football.

Metallurgy student = Badminton.
Nilesh = Basketball.
Deepak does not like Basketball, Tennis, Football, or Hockey.
The only remaining sport for Deepak is Cricket.
Therefore, Deepak likes Cricket (Option b).

Reference High-Resolution Figure / Slide:



Problem 07: Colour & Career Matrix Puzzle (`SEC01_MOD07_Q03`) ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

Origin Module: `7 _ Puzzle` | **Source Document:** `Puzzles.pdf`
Problem Statement:

```
SET 02: Seven friends P, F, R, T, Q, N and D study different specializations: IT, Civil, HRM, Marketing, Fir
```

Option Set:
- 1) P
- 2) T
- 3) F
- 4) Cannot be determined
- 5) None of these

Exact Answer: `2) T`

Complete Mathematical / Logical Deduction:

Given:
F = Girl, HRM, White.
P = Boy, Marketing, Red.
Q = Girl, IT, likes Yellow (since not Black/Green/Blue/White/Red).
The person studying Civil likes Blue.
Matching with R, T, N, D:
T likes Blue and studies Civil engineering.
Therefore, T is studying Civil Engineering.

Reference High-Resolution Figure / Slide:

